



**Graduate Studies
Masters Proposal Form**

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Major: EECE
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Thesis / Project Title: Deep Comparative Evaluation of Differential Privacy Integration
Across Multi-Phase Retrieval Augmented Generation Pipelines

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Tentative Dates For
Comprehensive Exam: 10/12/2025
Thesis Defense: 4/5/2026
Graduation: Spring

Date Submitted: 23/10/2025
Revised Date: _____

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**AMERICAN
UNIVERSITY^{OF} BEIRUT**

**MAROUN SEMAAN FACULTY OF
ENGINEERING & ARCHITECTURE**

Department of Electrical and Computer Engineering

Thesis Title

**Thesis Proposal
Fall 2022-2023**

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Submitted on: Date 23/10/2025

ACKNOWLEDGMENTS

I express my sincere gratitude to my advisors, committee members, and colleagues for their guidance and support throughout this investigation.

I am also grateful to my family and friends who have supported me.

ABSTRACT

Large Language Models (LLM) has emerged as a high-quality generative model. Pre-training on large-scale data which raises privacy concerns when it comes to sensitive domains. This concern has been considered by training the LLM using a stochastic gradient descent. Pre-training or fine-tuning on LLM requires massive computational power each time needed to keep model up-to-date or to be domain specific. Retrieval-Augmented Generation (RAG) has solved the problem of domain specificity and up to date, while personal data might be leaked during both retrieval and generation stage.

This thesis introduces a systematic framework for integrating Differential Privacy (DP) into each stage of a RAG system, enabling phase-wise evaluation of privacy–utility trade-offs and uncovering new insights into optimal DP deployment strategies. Specifically, it investigates the impact of various DP mechanisms applied at multiple points in the RAG pipeline, including the database level, pre-generation, and output stages, under both local and cloud-based LLM configurations. The framework is experimentally validated using two representative datasets, Enron Email and HealthcareMagic-101, through a comprehensive assessment of privacy and utility metrics.

Furthermore, this paper provides a quantifiable method for applying privacy on unstructured data, by identifying the optimal DP mechanisms and a guideline for local and cloud deployment. Finally, the paper aims to provide an RAG system that balances privacy and utility

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ABBREVIATIONS

LLM	Large Language Model
RAG	Retrieval-Augmented Generation
DP	Differential Privacy
PII	Personally Identifiable Information

CHAPTER 1

INTRODUCTION

1.1 Overview

Large language models (LLMs) represent a turning point in the history of artificial intelligence (AI) and natural language processing (NLP) [1]. LLM architectures can be categorized into three: encoder-only models, which embed language with rich contextual knowledge; decoder-only models, which are well suited for auto-regressive text generation [2]; and encoder-decoder-transformer-based models, which achieve state-of-the-art performance in complex reasoning, translation, conversational dialog, and fluent text composition [1]. These capabilities are due to extensive pre-training on diverse corpora [3].

LLMs also have notable disadvantages, two of the most significant being lack of up-to-date knowledge and hallucination. Hallucinations are due to biases in the training data, as well as the probabilistic nature of text generation, leading to inaccurate or unsupported output [4], [5]. This issue is particularly significant in the medical or law domains where the precision of the facts is important [6]. Thus, It emerges the need to have a balance system for privacy and utility. Retrieval-augmented generation (RAG) not only addresses the challenges of outdated information and hallucinations, but also enables the integration of domain-specific data [7], [8], [9]. RAG systems consist of an LLM (generator) with an external retriever database that fetches documents relevant to the user’s query [10], [11]. Integration of domain-specific data where accurate and reliable answers are essential, such as healthcare [12], legal advisory [13], and finance [14], introduces new privacy concerns that include the leakage of this information to unauthorized parties [15], [16], [17].

Adversaries used to prompt the retrieval in a way that led to the retrieval and the generator to extract and retrieve private data from the database [5]. This concern takes place even when the LLMs itself was trained on private data, and it reaches its exacerbated in case having a Cloud-hosted LLMs which can monitor or log prompts and retrieved text [5], [11].

Traditional mitigation strategies for the privacy of the RAG have limitations. To avoid fetching sensitive data pre-retrieval methods such as strict similarity thresholds or query filtering; To remove private information before generation post-retrieval techniques are applied, for example, re-ranking, summarization, or prompt-based instructions attempt to mask or remove private details before generation, leading to reduce obvious leaks, but

sensitive information still can be retrieved or generated despite these protections [11], [15]. Anderson et al. (2025) [15] propose an instruction-based defense that offers partial protection in three models: Flan, LLaMA, and Mistral. Here, LLaMA model that has system protection gets the focus in particular, leading to notably strong results in this setting. This highlights an important takeaway: while such defenses can be effective in specific cases, there remains a pressing need for more robust, general, and principled privacy-preserving mechanisms in RAG.

DP is used to limit the effect of the presence or absence of a record on the output based on a mathematical framework that protects the data by injecting carefully calibrated noise [18]. An example of applying DP in structure tasks is model training (e.g., DP-SGD) and database queries (via Laplace or Gaussian mechanisms) [19]. However, extending DP to unstructured text in a RAG pipeline is a challenge.

Recent research has explored differentially private mechanisms for unstructured data, such as text. DP-BART, a privatized rewriting model that trained in local DP by adding calibrated noise during the rewriting process that leads to text that preserves not only the meaning of the text, but also the privacy guarantees provided by the system [20].

Thus, RAG is a solution for some limitation in LLM such as hallucination, up-to-date data, and domain-specific. but it introduces privacy vulnerabilities that have been solved partially using the existing techniques. This proposal focuses on a core question:

How can DP-driven text rewriting be systematically integrated into RAG systems to mitigate sensitive information leakage, while maintaining retrieval accuracy and generation quality?

In this study a comprehensive comparison was performed to find the best stage for integrating DP in the RAG pipeline, providing a complete analysis of the trade-offs between utility preservation and privacy protection at each stage

1.2 Problem Statement and Motivation

1.2.1 *Motivation*

The motivation of this research is due to several observations and gaps which are:

- RAG provides a mechanism for grounding LLM in external knowledge, many of these domains have sensitive data such as healthcare, law, and finance, which increase the needs for privacy.
- traditional solutions are available, such as removing obvious identifiers, raising retrieval thresholds, paraphrasing responses, or adding instruction to prompts, but

do not protect data from being leaked [11], [15]. For example, applying paraphrasing or summarizing on the retrieved text does not remove all private information. Thus, it is necessary to have quantifiable privacy guarantees.

- DP provides a quantifiable privacy budget (ϵ), leading to upper limits of the risk of data leakage. Moreover, while a recent master’s thesis [1] identified potential points to integrate DP within the RAG pipeline, to our knowledge, no prior research has conducted a comprehensive stage-wise evaluation of DP-based rewriting strategies in all stages of the RAG process. This work advances the state of the art by systematically quantifying how each integration point affects the balance of privacy and utility, thus identifying optimal configurations that preserve model usefulness while ensuring strong privacy guarantees, which constitutes the core contribution of this study.

1.2.2 *Problem Statement*

How can preserve retrieval accuracy and generation quality while applying differential privacy using text rewriting approaches into different stage of RAG pipeline to mitigate the leakage of sensitive information?

1.3 Research Objectives

This research aims to address the following objectives:

- Identifying the impact of DP in Different Pipeline Stages, which are before retrieval, after retrieval (on retrieved documents before they go into the prompt), and after generation (on the model’s output), By comparing DP in different stages, the study will provide the best privacy protection and what are the trade-offs in terms of system utility. In parallel, the study will compare the effectiveness of traditional solutions such as adding instruction with stage-specific DP adaptations in RAG.
- Compare various DP text rewriting methods in balancing privacy vs. semantic fidelity by measuring metrics like retrieval recall, answer correctness, fluency, and privacy leakage for each method.
- To address the absence of standardized methods for assessing privacy leakage in textual data, this study aims to go beyond the theoretical guarantees of DP by conducting a comparative analysis and implementing practical evaluation techniques, including similarity-based leakage measures and LLM-based adversarial probing. The optimistic goal is to establish an empirical basis for quantifying privacy–utility

trade-offs in DP-based text rewriting, thus guiding the development of more privacy-preserving and reliable RAG systems.

- Study how the above outcomes differ between local RAG deployments where the threat might be through prompting and cloud-based RAG services where additional threats include network interception and cloud provider log [5]. Thus, as a hypothesis to this, a cloud scenario requires to enforce DP rewriting earlier before sending data to cloud, while in local scenarios, it depends on the privacy-utility needs

By answering the above stated research objectives, this work not only bridges the gap of evaluating privacy-utility trade-offs in integrating DP into different RAG pipeline, but it also provides insight of applying DP in real system while balance between privacy and utility

1.4 Research Scope and Structure

This thesis aims to compare the application of DP through rewriting text in different stages of the RAG pipeline, where this pipeline has two main stages, retrieval and generation. The generation stage can employ various LLMs such as the LLaMA, GPT, and Flan families to generate text based on the data retrieved from the database. To achieve the intended contribution, which is based on evaluating the privacy-utility trade-offs after integrating DP across several stages of the RAG pipeline and providing practical guidelines for real-world implementation, the following structure is proposed:

- **Chapter 2 – Literature Review:** It begins with introducing research on the RAG foundation and trustworthiness systems. Then it discusses the privacy and security risks in the RAG pipeline. After that, an existing mitigation is introduced, moving to present an overview of differential privacy on text. Finally, a master thesis of the Master’s thesis TUM has implemented a differential privacy database in the RAG pipeline.
- **Chapter 3 – Contributions:** List the contributions that consist of comparing DP at different stages and the formulation of guidelines for various deployment contexts.
- **Chapter 4 – Methodology:** Describes the experimental design, the datasets used (Enron Email and HealthcareMagic-101), the evaluation metrics, and the differential privacy mechanisms used throughout the study.

Thus, examination of DP at different RAG stages, increasing the body of work to have a secure and trustworthiness system. It not only introduces a practical methodology for implementing DP-RAG but also studies the theoretical insight of preserving privacy in unstructured generative contexts.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

In this chapter, a comprehensive review of the literature on RAG systems is provided. It starts first by introducing the foundation, challenges and trustworthiness of RAG system, then the classical RAG approaches and their evolution. Then it discusses the privacy and security risks in the RAG pipeline. After that, an existing mitigation is introduced, moving to present an overview of differential privacy on text. Finally, a master thesis of the Master’s thesis TUM has implemented a differential privacy database in the RAG pipeline.

2.2 Retrieval-Augmented Generation: Foundations, Challenges, and Trustworthy Systems

RAG is a framework used to enhance LLM, factuality, and domain specificity. Traditional LLM rely on pre-trained knowledge. time [7].

A RAG system that improves the factual accuracy of LLM in domain-specific and time-sensitive queries. Although the performance of the system is limited by the size and bias of the dataset, it is also limited by the reduced fluency of the fine-tuned 7B LLaMA-2 model [7].

Building on this, research on RAG introduces it as a mitigation for limitations of LLMs, such as hallucinations and outdated knowledge. It introduces that domain-specific applications for RA-LLMs is medicine, law, finance, and scientific discovery. One of the mentioned examples is the integration of scientific literature to support molecular discovery by MolReGPT, which reduces the risk of generating misleading hypotheses without empirical grounding. However, it highlights the future works for RAG system where trustworthiness system is one of them which is needed due to the malicious that affected humans where many RA-LLM systems rely on private or proprietary databases, making privacy protection an essential which is one of characteristics for trustworthiness. This in turn ensures both factual reliability and the security of sensitive data [10].

In line with this motivation, a comprehensive knowledge centric survey of RAG, which introduces its foundations, methods, applications, and evaluation practices. In addition,

It aims to identify emerging challenges, such as domain adaptation, efficiency, trustworthiness, and privacy, and outline future research directions. It shows that RAG has a diverse application across science, finance, education, healthcare, law, and industry, for example, in science it accelerates discovery, and in finance it improves business insights. It also highlights as a future work the need to have a RAG system that ensures trustworthiness [4].

Extending this discussion, a comprehensive survey has introduced a trustworthiness of RAG systems that consists of six dimensions, which are factuality, robustness, fairness, transparency, accountability, and privacy. It finds that proprietary instruction-aligned models perform most consistently, but across all the systems, privacy remains weak even for those that achieve the best score which are Baichuan2-7b-chat (26.0) and Llama2-13b-chat (22.0). This underscores the need to have a DP system to use proprietary instruction-aligned models with high privacy [3].

Table 2.1: Comparison of LLM Models by Strength, Weakness, and Privacy Score

Model	Type	Best Strength	Worst Weakness	Privacy Score
GPT-3.5-turbo	Closed	Transparency (61.2)	Privacy (0.0)	0
GPT-4o	Closed	Accountability (77.6)	Privacy (4.0)	4
Llama2-7b	Open	Factuality (14.0)	Privacy (0.0)	0
Llama2-7b-chat	Open	Transparency (29.5)	Factuality (0.0)	0
Llama2-13b	Open	Transparency (7.3)	Privacy (0.0)	0
Llama2-13b-chat	Open	Accountability (41.5)	Factuality (4.0)	22
Baichuan2-7b-chat	Open	Fairness (44.0)	Accountability (2.7)	26
Baichuan2-13b-chat	Open	Transparency (42.0)	Privacy (2.0)	2
Qwen2-7b-instruct	Open	Transparency (58.9)	Worst (2.0)	2
GLM-4-9b-chat	Open	Accountability (50.6)	Privacy (0.0)	0

The table shows a comparison of open and closed LLM models based on the six dimensions of the trustworthiness system, which are factuality, robustness, fairness, transparency, accountability, and privacy. It shows the main strengths and weaknesses of each LLM based on the trustworthiness dimensions. In addition, it shows the privacy score of each model to highlight the need to have a privacy awareness system.

2.3 Privacy and Security Risks in RAG Systems

As privacy and security are central concerns, a study reviews risks across RAG pipelines, from data ingestion, embedding, to retrieval and generation, and their mitigation. The threats addressed in this study are data leakage, embedding inversion, membership inference, prompt disclosure, indirect prompt injection, and jailbreak attacks, all of which lead to compromise in confidentiality, integrity, and trustworthiness. Each risk has been

mapped to specific mitigations such as anonymization, pseudonymization, access limitation, input validation, re-ranking, and exposure minimization. Several risks are executed as a black-box setting such as retrieval data leakage, membership inference attacks, knowledge corruption with black-box opinion manipulation, and indirect jailbreak attacks; where their mitigation includes anonymization, synthetic data generation, access limitation, input validation, re-ranking, distance thresholds, and summarization. However, jailbreak mitigation still needs further research. All of the mitigation strategies mentioned above were embedded in frameworks such as the NIST AI Risk Management Framework, the EU AI Act, and the OWASP guidelines [5].

Focusing more specifically on attacks, Zeng et al. (2024) investigate two questions which focus on privacy risk in RAG, and large language model (LLM) training data. Introduces prompting attacks, consisting of information-seeking and explicit command prompts. For LLM the following models are tested, which are Llama2-7b-Chat, Llama2-13b-Chat, and GPT-3.5-turbo which reproduce content at high rates. For instance, in untargeted attacks using the Enron email dataset, GPT-3.5-turbo reproduced 122 exact text matches and 208 near-matches out of 250 prompts, while in targeted attacks, it extracted 205 personally identifiable information (PII) instances. Although RAG introduces leak risk, it reduces the tendency to recall LLM during training. Mitigation strategies are evaluated which include re-ranking, summarization, and distance thresholds. This evaluation shows that summarization reduces the impact of untargeted leaks by 50%, but it has limited effect against targeted attacks. In addition to that, re-ranking showed negligible mitigation effects, and finally strict distance thresholds which show privacy-utility tradeoffs [11].

In addition to risk based on prompting, Anderson, Amit, and Goldstein (2025) an investigation of membership inference attacks (MIAs) that prompt the system using black-box and gray-box settings in three generative models (Flan-UL2, LLaMA-3-8B, and Mistral-7B) and two benchmark datasets (HealthCareMagic medical Q&A and Enron emails, each with 10,000 samples). As a result, the system shows in gray-box settings an average of AUC-ROC of 0.90, and it is nearly perfect for the Flan model (0.99), while in black-box it remained around 0.80. A mitigation strategy has been proposed by explicitly inserting an instruction into the RAG template, which reduces the attack for LLaMA and Mistral by up to 0.45 in AUC-ROC for gray-box settings, while it was ineffective against Flan. Thus, more robust approaches are required, such as DP mechanisms or adversarial awareness training, to provide comprehensive protection [15].

In addition to this, a membership inference attack (MIA) has been introduced to expose privacy vulnerabilities in RAG systems. It is a S2MIA that has semantic similarity between a target sample and the text generated by the RAG model, supplemented

with perplexity scores. Two methods were proposed, one that uses threshold-based classification and S2MIA-M, and one that supervised machine learning on similarity and perplexity features. The two methods were tested on five models LLaMA-2, Vicuna, Alpaca, and GPT-3.5, S2MIA consistently outperformed five existing MIA baselines that achieved ROC AUC scores up to 0.892, While S2MIA-M has limitations where it requires repeated querying, and the measure for privacy leakage has shown an effective result using BLEU which is sensitive to paraphrasing. Finally, this attack is a black box that does not need any knowledge of the system [21].

2.4 Existing Mitigation Strategies

To counter these forms of leakage, a Synthetic Attribute-based Generation with agent-based iterative system has been proposed which consists of two frameworks where in the first stage attributes extraction is then a generation method for key information and finally synthetic data is generated . This data is then moved to the next stage where an agent-based iterative is used, where one is used to assets privacy, and the other is to rewrite agent in to remove personally identifiable information (PII) or sensitive data. It uses data such as medical dialogu (HealthCareMagic-101) and a mixed Wiki-PII. For example, on HealthCareMagic, SAGE has score 0.113 which outperforms the paraphrasing that achieves ROUGE-L of 0.0943, the same for open-domain QA benchmarks (NQ, TQA, WQ, CT). SAGE has shown that it reduces the targeted and untargeted attack simulations to zero. Thus, this method has a limit that has been validated on healthcare and mixed-domain datasets, leaving its generalization to other underexplored domains .In addition, it only defends against targeted and untargeted extraction, attacks but does not provide formal guaranty, which still leaves residual risks. Finally, this generation process is performed using large language models (GPT-3.5/4), leading to high cost and reproducibility in resource-limited settings [6].

2.5 Differential Privacy for Unstructured data Text

The Differential Privacy (DP) formula provides a mathematical framework to quantify information leakage. This method ensures that the inclusion or exclusion of any single data can be bounded by a privacy budget ϵ . Although this was developed for structured numerical data, it was adapted to apply to text during LLM training [18].

DP has been used in deep learning through the DP-SGD (differentially Private Stochastic Gradient Descent) algorithm, this is done by injecting calibrated noise into gradient updates to protect training data. Thus, it provides a guaranty to unstructured [19].

DP has been applied under local DP by adding noise during paraphrasing and preserving the semantic meaning [20]. In addition to that, a DP-MLM and a 1-Diffractor where both apply DP by employing exponential mechanisms to privatize token-level prediction [8], [9].

2.6 TUM Master’s Thesis and RAG Privacy Process Framework

A master’s thesis by Bodea (2025) [1] has built an RAG privacy process diagram, which maps vulnerabilities such as dataset leakage, prompt injection and vector database risks to their corresponding mitigation strategies and shows at which stage each takes place and what to apply. Six mitigation methods have been applied in the data and DP methods, which are PII detection, PII deletion, PII labeling, 1-Diffractor, DP-Prompt, and DP-MLM. This was tested on two privacy-sensitive corpora, the BBC News and Enron Email datasets. As a result, a dependency on the data set and the method-specific has affected the result, where PII deletion offered the lowest privacy leakage in BBC with a score of 6, while DP-Prompt at $\epsilon = 150$ achieved the best privacy in Enron score of 9. In contrast, Diffractor with $\epsilon = 3$ provided the strongest semantic fidelity (highest ROUGE, BLEU, and cosine similarity) but also incurred severe leakage (45 on BBC, 74 on Enron). DP-MLM methods yielded scalable, balanced results, with coherence and fluency steadily improving as ϵ increased, although without surpassing Diffractor in utility or DP-Prompt in privacy. The limitations of the study include the dependence on a simplified RAG pipeline, a restricted set of techniques, applying DP at one stage, and the evaluation using LLM-as-a-Judge, which may introduce model-specific biases [1].

2.7 Research Gaps

This study introduces substantial progress for enhancing not only the privacy, but also the utility of RAG systems; however, several key challenges remain:

- Lack of formal privacy guaranty in most existing mitigation approaches.
- DP mechanisms have not yet been validated within the complete RAG pipeline, but rather in isolated text-rewriting tasks.
- currently, there is no standardized framework for quantifying privacy leakage in unstructured textual data.

- The trade-off between retrieval utility and privacy protection remains underexplored, particularly across different RAG deployment contexts (e.g., local vs. cloud-based environments).

This study addresses these gaps by presenting a comparative framework for applying DP at different stages of the RAG pipeline, coupled with privacy and utility assessments, thus establishing a foundational basis for balancing privacy preservation and system utility in RAG applications.

CHAPTER 3

CONTRIBUTIONS

3.1 Overview

This is the contribution of this research that aims to bridge that gap between RAG system and privacy preserving. The RAG system has enhanced the factuality and domain specificity of RAG system, it introduces privacy vulnerabilities due to the use of external database.

3.2 Systematic Integration of Differential Privacy in RAG:

An implementation of differentially private text rewriting will take place at multiple points in the RAG pipeline (database, document retrieval, and response generation). To our knowledge, this will be the first comparative work of applying DP guarantees in different stages of the RAG system on text, which will provide a blueprint for making RAG systems provably privacy-preserving. In addition to that, details about the mathematical privacy guaranty for each model such as a DP- masked language model (DP-MLM) which is a word substitution [8].

3.3 Comparative Evaluation of Privacy-Utility Trade-offs:

A comparative study of DP application strategies (database vs. document retrieval vs. response generation) and different rewriting techniques. Our evaluation will span multiple datasets such as HealthCareMagic-200k [22] and Enron-Email [23] using standard metrics such as the precision of the answers: and a comparative study will be included for different privacy metrics. By analyzing these results, conclusions will be drawn about which stage of the DP intervention offers the strongest privacy protection with the least utility cost.

3.4 Identification of Optimal DP Rewriting Techniques:

During the experiment, the most effective technique for applying DP-based text rewriting is expected to be identified or developed. The comparative study will focus not only on performance but also on computational efficiency and ease of integration. Based on the

obtained results, recommendations will be formulated for selecting the most suitable DP mechanisms for various processes.

3.5 Guidelines for Deployment Contexts:

Based on our comparative studies, a practical guideline will be provided to deploy the DP-RAG system in different settings. For cloud deployment, it may consider architecture where a rewrite of data takes place before sending them to the cloud. For an on-premises deployment, it may suggest applying DP at one or two points and quantifying the resource overhead. In addition, the choice of ϵ (privacy budget) depends on the different sensitivity of domains. Thus, this contribution results in real-world implementation considerations.

3.6 Foundations for Quantifying Privacy in RAG:

As an exploratory but important contribution, a formulating to measure information exposure might defined where deriving a closed-form expression is a challenging, an initial observation will be tackled even if a formal formula is left to be a future work.

In summary, this work aims to state the art in securing NLP data in RAG system using DP methods, which have mathematical framework for protecting individual information by injecting carefully calibrated noise. The aim is not only to improve privacy for current systems, but also to have a set of foundations for a new line of research at the intersection of privacy, text rewriting, and large-scale language model systems.

3.7 Implications and Impact

This research provides a blueprint for the deployment of a privacy RAG system for the private domain which has a not only theoretical significance but also practical. It is based on privacy compliance, such as GDPR. In addition to that, the study aims to have trustworthiness frameworks for the RAG system.

3.8 Summary

Table 3.1 provides an overview of privacy-preserving methods at different stages of RAG pipeline which are database (DB), pre-generation, and post-generation stages. The methods are classified based on their categories into PII detection and anonymization methods, traditional mitigation methods for RAG models, and differential privacy (DP) methods for text data, and finally a special section for DP extensions that are advanced. In this table (✓) means that this method will be applied at this stage, and (✗) represents that

this method has been applied at a stage in the reference paper. As shown in Table 3.1, the proposed contribution covers all stages of the RAG pipeline and integrates advanced privacy-preserving techniques, clearly highlighting the importance and distinct advantage of the proposed approach in comparison with existing solutions.

Table 3.1: This table shows a Privacy-Preserving Mechanisms, their Applicability Across RAG Pipeline Stages, and Related Work References

Method / Model	DB	Before Gen.	After Gen.	Reference / Source Work
PII Detection and Anonymization Methods				
PII Deletion	✓✗	✓	–	[1]
PII Labeling	✓✗	✓	–	[1]
PII Replacement (Synthetic Data)	✓✗	✓	–	[1]
Traditional Mitigation Methods for RAG Systems				
Instruction-Based Prompt Defense	–	✓✗	–	[15]
Summarization (with/without paraphrasing)	–	✓✗	–	[21] [11]
Re-ranking	–	✓✗	–	[21] [11]
Differential Privacy Mechanisms for Textual Data				
DP-MLM (Differentially Private Masked LM)	✓✗	✓	✓	[1]
1-Diffractor	✓✗	✓	✓	[1]
DP-Prompt	✓✗	✓	✓	[1]
DP-BART	✓	✓	✓	
Advanced Differential Privacy Extensions				
DP-MLM + PII Labeling	✓	✓	✓	Our contribution
DP-MLM Agent-Based System	✓	✓	✓	Our contribution .

✓= Applied in this paper; ✗= Applied in referenced studies;

CHAPTER 4

METHODOLOGY

4.1 Overview

This chapter outlines the research design, black-box attacks, mitigation strategies, RAG architecture, dataset, and evaluation methods employed in this study. The purpose of this study is to evaluate different methods for DP, not only in various stages of the RAG pipeline, but also by applying two different methods in different stages simultaneously. The stages are database, post-retrieval, and after generation. The methodology is designed to quantify the privacy–utility trade-offs.

The chapter proceeds as follows. Section 4.2 introduces details about the research framework and the different stages of RAG pipeline. In section 4.3 the attack that this study aims to mitigate is defined. Section 4.4 defines the differences in the mitigation studies that were used to conduct the comparative studies. Section 4.5 introduces the main component of the RAG system. Section 4.6 provides definitions and domain classifications for all datasets, which form the foundation of the comparative analysis. Finally, a set of matrices has been defined in section 4.7 to evaluate the retrieval process, the generation process, and privacy.

4.2 Research Design

A comparative analysis is applied to study how the application of DP in different stages of the RAG pipeline will affect the balance between privacy and utility. The core question of this study is:

How can differential privacy–driven text rewriting be systematically applied at different stages of a RAG pipeline to minimize sensitive information leakage while preserving retrieval accuracy and generative fluency?

Three primary configurations are tested:

1. **Dataset Differential Privacy:** Before storing data, DP-based rewriting is implemented.

2. **Post-Retrieval Differential Privacy:** Prior to getting grouped with the user query, DP-based rewriting is applied on retrieved documents
3. **After Generative-Stage Differential Privacy:** DP-based rewriting is applied at the output before sending it to the user.

At each stage, a comparative study will be performed against traditional methods, such as instruction-based defenses [15], PII deletion methods, and DP in text. The outcomes will be assessed not only on the bases of quantitative metrics (retrieval relevance, answer accuracy, privacy leakage) but also on qualitative evaluation (semantic fidelity and fluency).

4.3 Black-Box Attacks

Integrating the external retrieval into the LLM using the RAG system has led to the introduction of several attacks against sensitive information. It's true that these attacks vary in objectives and methods, but they have in common the goal of compromising privacy.

The attack prompt can be represented as:

$$q_{\text{attack}} = \underbrace{\text{information}}_{\text{targeted content}} + \underbrace{\text{command}}_{\text{instruction to LLM}} \quad (4.1)$$

where:

- **Information** is the content that the adversary aims to retrieve
- **command** is used to instruct the model to output the retrieved data

4.3.1 Retrieval Data Leakage

Data leak is categorized as targeted and untargeted attack. When the adversary aims to retrieve specific personal data, it is called targeted attack. When the adversary seeks to obtain the utmost amount of personal information, it is called untargeted attack. When manipulating prompts, the attack becomes a black-box which requires both retrieving the needed data and letting the LLM generate it. [5].

4.3.2 Membership Inference Attack

It enables the adversary to know if there is a specific user in the database. To estimate the membership function $M(x)$, this attack was applied as a black-box.

$$M(x) = \begin{cases} 1, & \text{if } x \text{ in the dataset,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.2)$$

Successful attacks result in identifying that the data has been used without necessarily exposing the data itself [5].

4.3.3 *Retrieval Data Disclosure during Prompting*

This occurs in the retrieval stage when sensitive information is retrieved during the prompting stage. In which the system retrieves sensitive information that will be generated based on the user’s prompt. This risk is relevant when the LLM is located in the cloud or when you have hybrid-cloud environments that allow third parties to access the forwarded documents [5].

Hence, these attacks reveal that RAG systems are prone to risks at various stages: retrieval, pre-sending to LLM on a cloud or hybrid-cloud, and throughout generation. This pushes to apply Differential Privacy at different stages of the RAG pipeline with the bid of reducing data leaks.

4.4 Mitigation strategies for Black-Box Attack

4.4.1 *PII Detection:*

- **PII Deletion:** It is a data anonymization approach that uses Microsoft Presidio’s Analyzer to scan for personally identifiable information (names, locations, or organizations) using, for example, named-entity recognition (NER). This is done based on a confidence threshold of 0.5. although this method is effective, it loses its contextual information.
- **PII Labeling:** Detecting personal data- instead of deleting them- comes first with PII Labeling. It is substituted by a suitable label like <PERSON>, <LOCATION>, or <DATE_TIME>. This way it maintains the syntactic coherence of the text.
- **PII Replacement with Synthetic Data:** Instead of replacing the detected entity with a place holder, it replaces using a LLM model by forwarding a prompt to generate synthetic data the will not only maintain the syntactic structure but also keeps the text realistic
- **PII Replacement with DP-MLM:** This aims to replace the detected entity with a mask token to generate synthetic data that are differentially private, and in

the same time save grammatical structure and realistic of text. This will be more explored in section 4.4.3.

4.4.2 *Traditional Mitigation Method for RAG System:*

- **Instruction-Based Prompt Defense:** This defense is achieved by defining a prompt template that controls how LLM handle the input data, guiding their behavior to forbid generating sensitive data.
- **Summarization:** This method applies summarization to the retrieved data before the generation stage. It comprises two types: The first type is Extractive Summarization, which retrieves the original sentences without any changes. The second type is Abstractive summarization, which involves paraphrasing or rewriting sentences
- **Re-ranking** it is used for optimization purpose, but then it test its ability to mitigate against data leak by scoring the relevance of each retrieved document. This way, we stress the retrieved data that comply with the query. Traditional mitigation solutions use it this way. A differentially private scoring method ameliorates this method.

4.4.3 *Differential Privacy Mechanisms for Textual Data*

- **DP-MLM** is a differentially private masked language model. Mask means that it is based on BERT-based masked language models. This model rewrites text token by token, which masks and resamples words based on temperature-based sampling (randomness attributes for the LLM models) with a logit clipping in order to have privacy constraints. The privacy parameter ϵ responsible for controlling the balance between privacy and semantic fidelity. Coherence has been improved by concatenating the original sentence with the mask one [8]. The model will anticipate that likelihood that the mask token is the word v of a vocabulary $\in V$, where the masked token is located in position l in a sentence:

$$P[w_l = v] = P[v \mid w_1 \cdots w_{l-1}, w_{l+1} \cdots w_n] \quad (4.3)$$

Which is based on exponential mechanism that generate the output by persevering the privacy:

$$P[M(C_l) = v_i] \propto \exp \left(\frac{u(C_l, v_i)}{T} \right) \quad (4.4)$$

– C_l it represents the location of the mask token

- $u(C_l, v_i)$ represent the utility function, which in this case is the logit that the mask token C_l is the token $v_i \in V$
- T it represents the temperature parameter that controls randomness, and ε it represents the privacy budget parameter.

$$T = \frac{2\Delta u}{\varepsilon}, \quad (4.5)$$

Δu represented as :

$$\Delta u = \max_{v_i \in V} \max_{C_l, C'_l \text{ adjacent}} |u(C_l, v_i) - u(C'_l, v_i)| \quad (4.6)$$

This mechanism replicates repeats the process token by token in a sentence that comprises a sequence of n where this process lies under the sequential composition rule

$$\varepsilon_{\text{total}} = n \cdot \varepsilon \quad (4.7)$$

- **D-MLMP with PII Labeling:** To decrease privacy budget, Microsoft Presidio’s Analyzer is used. The latter is responsible for detecting PII (personally identifiable information) including names, dates and locations. Then a differential privacy mask language model is used to decrease the aggregate privacy budget to be:

$$\varepsilon_{\text{total}} = n_{\text{sensitive}} \cdot \varepsilon \quad (4.8)$$

It’s true that the traditional PII methods that detected sensitive information maintains the grammatical structure; yet, this process preserves not only the grammar and contextual coherence but also differential privacy.

- **DP-MLM Agent-Based System:** A multi-agent system is designed to balance privacy and data utility. It consists of two stages:
 - First stage: This stage starts by identifying the key attributes such as medical information including symptoms, diagnoses, or treatment. Then, using a Large Language Model (LLM) to extract information that are related to the attributes. Finally, a privacy-preserving synthetic data generation is done using DP-MLM.
 - Second stage: This stage contains two specialized agents. The first agent detects if there is a privacy risk such as personally identifiable information (PII). This is detected using Microsoft Presidio’s Analyzer, but the extend

here is to recognize inferable contextual information that may compromise privacy. If no private data selected the process ends here, else the data is sent to the second agent that replace the detect private information using a DP-MLM. This cycle is iterative until no private information remains. Thus, the total privacy loss is proportional to the number of sensitive tokens identified and processed by the system

$$\varepsilon_{\text{total}} = n_{\text{sensitive}} \cdot \varepsilon \quad (4.9)$$

- **1-Diffractor** It applies differential privacy by ranking a list of semantically similar words which are generated by a word embedding model such as GloVe, then, **NLTK** used to differ between stopwords, punctuation, and content token. Then, content token is replaced by another token from the precalculated semantic list. Thus, the process selected the closer word instead of more likely. It consists of two hyperparameter Epsilon which controls privacy-utility trade-off, and Gamma which determine the size of the list of tokens. Each word w assign an index such that the distance between two word is defined as:

$$d_V(w, w') = |\Phi(w) - \Phi(w')|. \quad (4.10)$$

Then a privatize token is output by adding noise to the index such as Laplace $G(0, \frac{1}{\varepsilon})$ which ensure ε -differential privacy. A truncated function is used to keep index in the range.

After that a decoding is applied in order to get the token using Φ^{-1} . Finally, each token is perturbed independently which lead to has a the following composition rule for n token:

$$\varepsilon_{\text{total}} = n \cdot \varepsilon. \quad (4.11)$$

- **DP-Prompt** It is based on transformer architecture that generates text by injecting noise into the decoding part of the model. The process starts by tokenizing the input text and formatting it as : "Document: original_text. Paraphrase of the document:" Now the model clips the logit to its output probabilities to be limited in the range min_logit and max_logit where the sensitivity function is :

$$\text{sensitivity} = \text{max_logit} - \text{min_logit} \quad (4.12)$$

Then, the temperature parameter need to be calculated based based on the sensi-

tivity function:

$$\text{temperature} = \frac{2 \times \text{sensitivity}}{\epsilon} \quad (4.13)$$

To control the privacy budget, epsilon is the hyperparameter used. Lower value reflect higher privacy while higher value reflects higher quality.

- **DP-BART** it pre-train on public data where its architecture made of an encoder-decoder model. A matrix z is output by the model which belongs to $\mathbb{R}^{\ell \times d_{\text{tok}}}$ for an input x . Then a clipping is applied to each vector:

$$\bar{z}_i = \min(\max(z_i, -C), C) \quad (4.14)$$

A noise is added using Laplace or Analytic-Gaussian to obtain new representation of the vector $\bar{z}_i$ to be $\dot{z}$

$$\dot{z} = \bar{z} + (Y_1, \dots, Y_n) \quad (4.15)$$

- it achieve $(\epsilon, 0)$ DP with Laplace and ℓ_1 Sensitivity

$$\Delta_1 f = \max_{x,y} \|f(x) - f(y)\|_1 = 2Cn \quad (4.16)$$

- it achieve (ϵ, δ) DP with the analytic Gaussian mechanism and ℓ_2 Sensitivity

$$\Delta_2 f = \max_{x,y} \|f(x) - f(y)\|_2 = 2C\sqrt{n} \quad (4.17)$$

Then, a rewritten text is generated

$$\hat{y} = \text{DEC}(\dot{z}) \quad (4.18)$$

This privatization is done in the inference time

It is improved during training by pruning neuron at same index, which required a few additional training in order not to degrade utility. This process is repeated until the target dimension is reached, then the weight and the pruned indices are frozen. Finally, the above step now applies in the inference time, using a pre-train model that reduces the sensitivity by reducing d_{tok} which in turn leads to reducing n to become $n' = \ell \cdot d'_{\text{tok}}$

In addition, this architecture improves the privacy–utility trade-off, but it has a list of limitations:

1. pruning may lead to loss of information.
2. It needs to be trained on public data, which may lead to domain mismatch
3. pruning the same neuron across all tokens leads to losing context features.

4.5 RAG Architecture

Retrieval-Augmented Generation (RAG) It is based on integrating external database, solving the problem of out-of-date information and domain-specific data in LLM.

4.5.1 *RAG Components*

1. **Retriever Component:** This stage retrieves external information based on the user query. Dense Passage Retrieval (DPR) is commonly used to obtain this information by encoding both the query and the documents into vector representations using an embedding model. This enables efficient similarity matching. This ensures that the LLM gets the most relevant document, leading to accurate data generation .
2. **Generator Component:** the retrieved data are concatenated with the user query and passed to a generative model such as GPT to produce the final response.

4.5.2 *Workflow of RAG*

The RAG process typically involves several steps:

- The user query is transformed into a vector representation.
- A vector database is searched to retrieve the most relevant documents.
- The retrieved documents are combined with the query {User Prompt} + {Retrieved Content} and then passed to the generative model.
- The model generates a response by blending its internal knowledge with the retrieved context.

4.6 Datasets

It is important in our comparative study to generalize over domains for these reasons that two publicly available datasets that contain sensitive information are used. privacy-sensitive datasets are used.

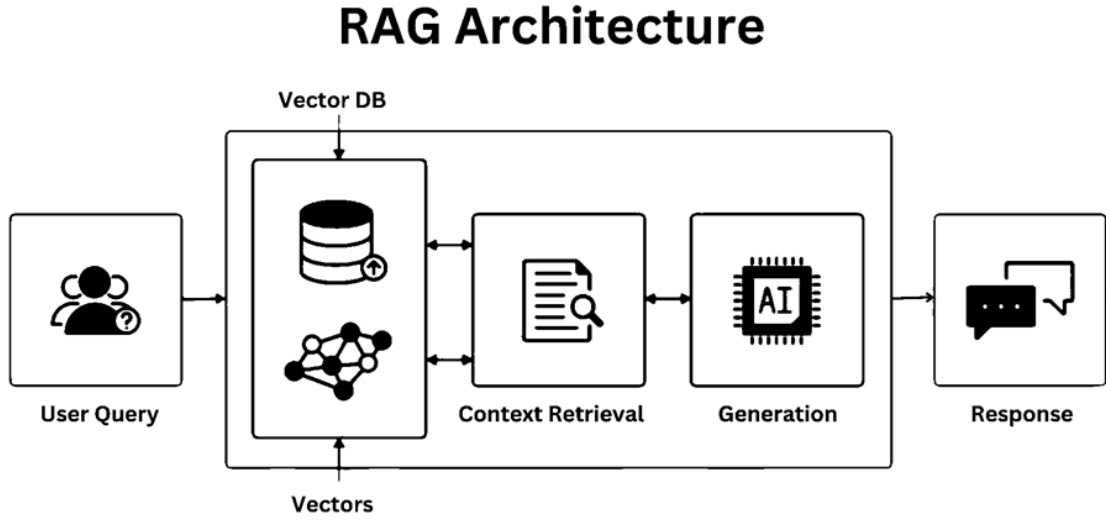


Figure 4.1: RAG Architecture Overview

4.6.1 *Enron Email Dataset*

This data set consists of approximately 500,000 internal emails that are exchanged among Enron employees. This data set has been used extensively in studies that tackle privacy leakage and membership inference attacks [1], [11], [15]. These data contain personally identifiable information (PII), including names, addresses, and business correspondence, which in turn makes it realistic to evaluate privacy vulnerabilities in RAG systems.

4.6.2 *HealthcareMagic-101 Dataset*

It covers about 200,000 pairs of doctor–patient dialogue [6], [15]. It is important to mention that these data contain sensitive information related to medical matters such as personal health details. That in turn increases the need to have a realistic evaluation of privacy risk in the RAG system

A pre-processing is necessary to prepare the data which might undergoes in the following procedure:

- Remove duplicates
- Normalize text encoding
- Split them into training, validation, and testing subsets
- A vector representation is created using embedding model

- Index documents in a FAISS-based vector databas.

4.7 Metrics for Evaluation

The RAG architecture comprises two components: information retrieval and a large language model used to generate data for which different metrics must be used to get evaluated.

4.7.1 Retrieval Component Metrics

This component is quantified by the relevance and proficiency of the retrieved information.

- **Precision@K:** it decides the performance of a ranking system by computing how many of the top- k retrieved documents are pertinent [24]

$$\text{Precision@K} = \frac{TP}{TP + FP} \quad (4.19)$$

- **Recall@K:** is used to estimate the precision of the extracted data by calculating the proportion of relevant documents among the top-results k . [24]

$$\text{Recall@K} = \frac{|RD \cap \text{Top}_k^d|}{|RD|} \quad (4.20)$$

- **F1 Score:** it is used to assign an equally significant for bot false positive and negative, by balancing between precision and recall. [24]

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.21)$$

- **Mean Reciprocal Rank (MRR):** It generates a rank list that reveals the importance of a pertinent document. [24]

$$MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i} \quad (4.22)$$

4.7.2 Generation Component Metrics

These benchmarks are used to evaluate the quality of the generative stage:

- **Exact Match (EM):** determine if the generated answer is match or not. [24]

$$EM = \begin{cases} 1 & \text{if output} = \text{reference}, \\ 0 & \text{otherwise.} \end{cases} \quad (4.23)$$

- **ROUGE:** It is suitable for long term generation since it measures the n -gram overlap between generated and reference responses. [1], [24]
- **BLEU (Bilingual Evaluation Understudy):** It finds the precision of n -gram between the generated and the reference text. [1], [24]:

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (4.24)$$

- **BERTScore:** it captures the linguistic relation by utilizing a contextual embedding from a pre-train BERT model [1], [24].

$$\text{BERTScore}(x, y) = \frac{1}{|x|} \sum_{i=1}^{|x|} \max_j \cos(\mathbf{h}_i^x, \mathbf{h}_j^y) \quad (4.25)$$

4.7.3 Privacy Evaluation Metrics

Since privacy risks are not considered in the traditional IR and NLG metrics, there is recourse to additional evaluation methods:

- **Semantic Similarity Leakage (SSL):** it Computes semantic overlap between privatized and original text, founded on cosine similarity over contextual embeddings
- **LLM as A Judge:** prompt or train the LLM to detect if there is any private information in the text

Together, these steps form an exhaustive framework for the evaluation of utility and privacy in DP-enhanced RAG systems.

4.8 Expected Outcomes

In this chapter, the expected outcomes will be studied.

4.8.1 *Comparative Study:*

A comparative will be applied on different methods that are used to protect privacy, which are PII detection, traditional mitigation, and differential privacy at different stages of the RAG pipeline

- Database
- After retrieval.
- After generation.

Thus, a clear vision of where and how to apply DP mechanism to reduce privacy leak while preserving accuracy

4.8.2 *Quantitative Evaluation Method for Privacy in RAG:*

A comparative study based on a quantifiable analysis to assess privacy in the RAG system. This comparative study aims to determine the most suitable approach to measure the effectiveness of DP in practice.

4.8.3 *Privacy-Utility Trade-offs:*

An evaluation using privacy and utility metrics that lead to a system that balances privacy and utility.

4.8.4 *Optimal DP Text Rewriting Techniques:*

Comparing different techniques will provide which mechanism maintains best semantic fidelity, which one is the best computational, and how each perform across deployment local vs. cloud.

4.8.5 *Deployment Guidelines:*

Provide guidelines to preserve privacy and utility.

- **Cloud-based deployments:** Apply DP before sending data to the cloud.
- **On-premises deployments:** DP can be applied at any stage based on privacy-utility needs.

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